

Problem 1

Part a). We take cp (the column player) to be **Player 1** with budget 3, and rp (the row player) to be **Player 2** with budget 4.

Pure strategies. Cp's strategies are all the ordered pairs of 2 numbers that sum to 3– that is, cp's budget. These are,

$$\text{cp strategies (columns): } C_1 = (0, 3), C_2 = (1, 2), C_3 = (2, 1), C_4 = (3, 0).$$

Similarly, rp's strategies are all the ordered pairs of 2 numbers that sum to 4– that is, rp's budget. These are,

$$\text{rp strategies (rows): } R_1 = (0, 4), R_2 = (1, 3), R_3 = (2, 2), R_4 = (3, 1), R_5 = (4, 0).$$

Payoff matrix A. Payoffs in the payoff matrix are described from the point of view of cp. Each entry is the sum over the two battlefields of $\pm v_i$ according to strict win/loss/tie. This yields the 5×4 matrix

$$A = \begin{pmatrix} -4 & -1 & -1 & -1 \\ -3 & -4 & -1 & -1 \\ 1 & -3 & -4 & -1 \\ 1 & 1 & -3 & -4 \\ 1 & 1 & 1 & -3 \end{pmatrix}.$$

LP for cp. Following the discussion seen in class and in the lecture notes, the strategy for the cp is given by the optimal solution to the following LP

$$\begin{aligned} \max \quad & v \\ \text{s.t.} \quad & -4x_1 - x_2 - x_3 - x_4 \geq v & (\text{row } R_1) \\ & -3x_1 - 4x_2 - x_3 - x_4 \geq v & (\text{row } R_2) \\ & x_1 - 3x_2 - 4x_3 - x_4 \geq v & (\text{row } R_3) \\ & x_1 + x_2 - 3x_3 - 4x_4 \geq v & (\text{row } R_4) \\ & x_1 + x_2 + x_3 - 3x_4 \geq v & (\text{row } R_5) \\ & x_1 + x_2 + x_3 + x_4 = 1, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned} \tag{1}$$

LP for rp. Similarly, the strategy for the rp is given by.

$$\begin{aligned} \min \quad & u \\ \text{s.t.} \quad & -4y_1 - 3y_2 + y_3 + y_4 + y_5 \leq u & (\text{column } C_1) \\ & -y_1 - 4y_2 - 3y_3 + y_4 + y_5 \leq u & (\text{column } C_2) \\ & -y_1 - y_2 - 4y_3 - 3y_4 + y_5 \leq u & (\text{column } C_3) \\ & -y_1 - y_2 - y_3 - 4y_4 - 3y_5 \leq u & (\text{column } C_4) \\ & y_1 + y_2 + y_3 + y_4 + y_5 = 1, \\ & y_1, y_2, y_3, y_4, y_5 \geq 0. \end{aligned} \tag{2}$$

Solution. Solving (1)–(2) with any LP solver returns optimal mixed strategies $(\bar{x}, \bar{y})$ given by:

$$\bar{x}^T = \left(\frac{27}{100}, \frac{9}{100}, \frac{39}{100}, \frac{1}{4} \right), \quad \bar{y}^T = \left(\frac{1}{4}, \frac{39}{100}, \frac{9}{100}, \frac{27}{100}, 0 \right),$$

and the value of the game (cp's expected payoff) is

$$\bar{v} = \bar{u} = -\frac{181}{100}.$$

Part b). Let R be all the ordered pairs of k nonnegative integers that sum to b_1 , and let C be all the ordered pairs of k nonnegative integers that sum to b_2 . In the payoff matrix A we have a row for each entry in $r = (r_1, r_2, \dots, r_k) \in R$ and a column for each $c = (c_1, c_2, \dots, c_k) \in C$. The entry in row r and column c of A is given by

$$A[r, c] = \sum_{1 \leq i \leq k: c_i > r_i} v_i - \sum_{1 \leq i \leq k: c_i < r_i} v_i.$$

The cp will solve the LP

$$\begin{aligned} \max \quad & v \\ \text{s.t.} \quad & \sum_{c \in C} A[r, c] x_c \geq v, & \forall r \in R, \\ & \sum_{c \in C} x_c = 1, \\ & x_c \geq 0, & \forall c \in C. \end{aligned}$$

and the rp will solve the LP

$$\begin{aligned} \min \quad & u \\ \text{s.t.} \quad & \sum_{r \in R} A[r, c] y_r \leq u, & \forall c \in C, \\ & \sum_{r \in R} y_r = 1, \\ & y_r \geq 0, & \forall r \in R. \end{aligned}$$

Problem 2

Rewrite (1) as

$$\begin{aligned} \max \quad & v \\ & v - [Ax]_R \leq 0 \\ & v - [Ax]_S \leq 0 \\ & v - [Ax]_P \leq 0 \\ & x_P + x_R + x_S = 1 \\ & x_P, x_R, x_S \geq 0 \end{aligned}$$

By letting y_R, y_S , and y_T the dual variables corresponding to the first three constraints and u the dual variable corresponding to the last constraint, we can write its dual

$$\begin{aligned} \min u \\ u - [A^T y]_T &\geq 0 \\ u - [A^T y]_R &\geq 0 \\ u - [A^T y]_S &\geq 0 \\ y_P + y_R + y_S &= 1 \\ y_P, y_R, y_S &\geq 0 \end{aligned}$$

which is, after rearranging, equal to (2). The result then follows by strong duality.

Problem 3

Solution. a) The dual is

$$\begin{aligned} \min 4y_1 + 6y_2 \\ \text{s.t. } y_1 + 2y_2 &\geq 3, \\ 2y_1 + y_2 &\geq 2, \\ y_1 + 3y_2 &\geq 1, \\ y_1, y_2 &\geq 0. \end{aligned}$$

Candidate $x^* = (\frac{8}{3}, \frac{2}{3}, 0)$. The two primal constraints are tight:

$$x_1 + 2x_2 + x_3 = \frac{8}{3} + \frac{4}{3} = 4, \quad 2x_1 + x_2 + 3x_3 = \frac{16}{3} + \frac{2}{3} = 6.$$

Complementary slackness for $x_1 > 0, x_2 > 0$ enforces equalities

$$y_1 + 2y_2 = 3, \quad 2y_1 + y_2 = 2,$$

whose solution is $(y_1, y_2) = (\frac{1}{3}, \frac{4}{3}) \geq 0$. The third dual inequality also holds: $y_1 + 3y_2 = \frac{1}{3} + 4 = \frac{13}{3} \geq 1$ and $y \geq 0$, so y is feasible for the dual. We are left to check the condition that $y_1 > 0, y_2 > 0$ impose on the primal constraints: they require both of them to be tight at x^* . Since this is the case, we conclude that x^* is optimal for the primal (and y for the dual).

Candidate $\tilde{x} = (3, 0, 0)$. Here $x_1 + 2x_2 + x_3 = 3 < 4$ (slack) and $2x_1 + x_2 + 3x_3 = 6$ (tight). CS forces $y_1 = 0$, while $x_1 > 0$ requires $y_1 + 2y_2 = 3$, which, together with $y_1 = 0$, implies $2y_2 = 3$, thus $y_2 = \frac{3}{2}$. But then $2y_1 + y_2 = \frac{3}{2} \not\geq 2$, violating the dual constraint. Hence there is no feasible dual solution that satisfies the complementary slackness with $\tilde{x}$, which is therefore not optimal.

b) The dual is

$$\begin{aligned} \min 3y_1 + 2y_2 \\ \text{s.t. } y_1 &= 2 \quad (x_1), \\ y_2 &= 1 \quad (x_2), \\ y_1 - y_2 &= 1 \quad (x_3), \\ y_1, y_2 &\geq 0. \end{aligned}$$

Thus, the dual has the only solution $(y_1, y_2) = (2, 1)$. Note that, since all primal variables are free, and $y_1, y_2 > 0$, a solution to the primal is optimal if and only if it is feasible and it satisfies at equality both constraints.

Candidate $x^* = (2, 3, 1)$. Both primal constraints are tight: $x_1^* + x_3^* = 3$ and $x_2^* - x_3^* = 2$. Thus, x^* is optimal.

Candidate $\tilde{x} = (0, 0, 0)$. None of the primal constraints are slack. Thus, $\tilde{x}$ is not optimal.

Problem 4

a) See attached code.

b) See attached code.

c) Yes. Indeed, we have seen in class that (M, p) form an equilibrium if and only if $(M, (p, u))$ satisfy the complementary slackness (for some u) for the pair of primal / dual LPs where the primal is the maximum weighted matching problem from the body of the exercise, and the dual is its dual. Since every optimal primal/dual pair satisfies the complementary slackness, $(\overline{M}, p^*)$ satisfy the complementary slackness, and it is in particular an equilibrium.